

1. Sutherland 5.14

2. Sutherland 5.15

3. Sutherland 6.3 Modified.

Show that if X is a metric space and $p \in X$, then the singleton $\{p\}$ is closed. Then show in one line that any finite subset of X is also closed.

4. Sutherland 6.6

5. **(AI Free)** Suppose F is closed in X and G is closed in Y . Show that $F \times G$ is closed in $X \times Y$. You may find it nice to do this quickly using sequence arguments.

6. **AI Use.** Please describe specifically how you used AI to help complete this assignment.